74LS00 (IC4), decoder HT12D (IC5), two timers NE555 (IC6 and IC9), shift register 74LS164 (IC7), two transistors BC547 (T1 and T2), transistor SL100 (T3), 12V relay and a few discrete components.

Transformer X2 converts 230V AC to 12V-0-12V, which is rectified by the diodes (D5 and D6) and regulated by IC8 to provide regulated 5V. Capacitors C3 and C4 are used for filtering. 5V supply enables the circuit while 12V goes to relay circuit via D3.

RX1, a 433MHz UHF RF receiver module, is used to receive and demodulate the ASK modulated RF signal transmitted by TX1 module of the transmitter unit placed near the overhead tank. Demodulated output is a train of rectangular pulses representing the status (logic levels) of twelve (eight address and four data) bits of the encoder at the transmitting end.

Transistor T1 is used as a pulse amplifier to amplify the signal output from RX1. It raises pulse height to CMOS-compatible logic-1 level (>3.5V at 5V VDD). This compatible output is then fed to pins 12 and 13 of IC3 (internal NAND gate N4). NAND gate helps get pulses of perfect rectangular-wave shape.

Output (pin 11) of NAND gate N4 is fed to input (pin 14) of $2^{12}$ special decoder IC5. The decoder accepts and decodes the signal from gate N4 when, and only when, logic levels of its 8-bit address match with those of the encoder at the transmitter end. Since address bits of the encoder are not fixed and change with changes in water level in the tank, a special address scanning system is used. The system continuously feeds all possible address bits to the decoder until a match is found.

The address scanning system comprises a Johnson counter constructed around an 8-bit serial-in-parallel-out shift register 74LS164 (IC7). A clock pulse generator NE555 (IC6) feeds the required clock pulse to the shift register continuously. So the counter advances and its outputs follow sequentially the logic levels tabulated in Table 1, first in forward and then in reverse order. IC6 operates in astable mode. Clock pulse rate is governed by resistors R12 and R15, and capacitor C7. LED11 blinks to show the clock's

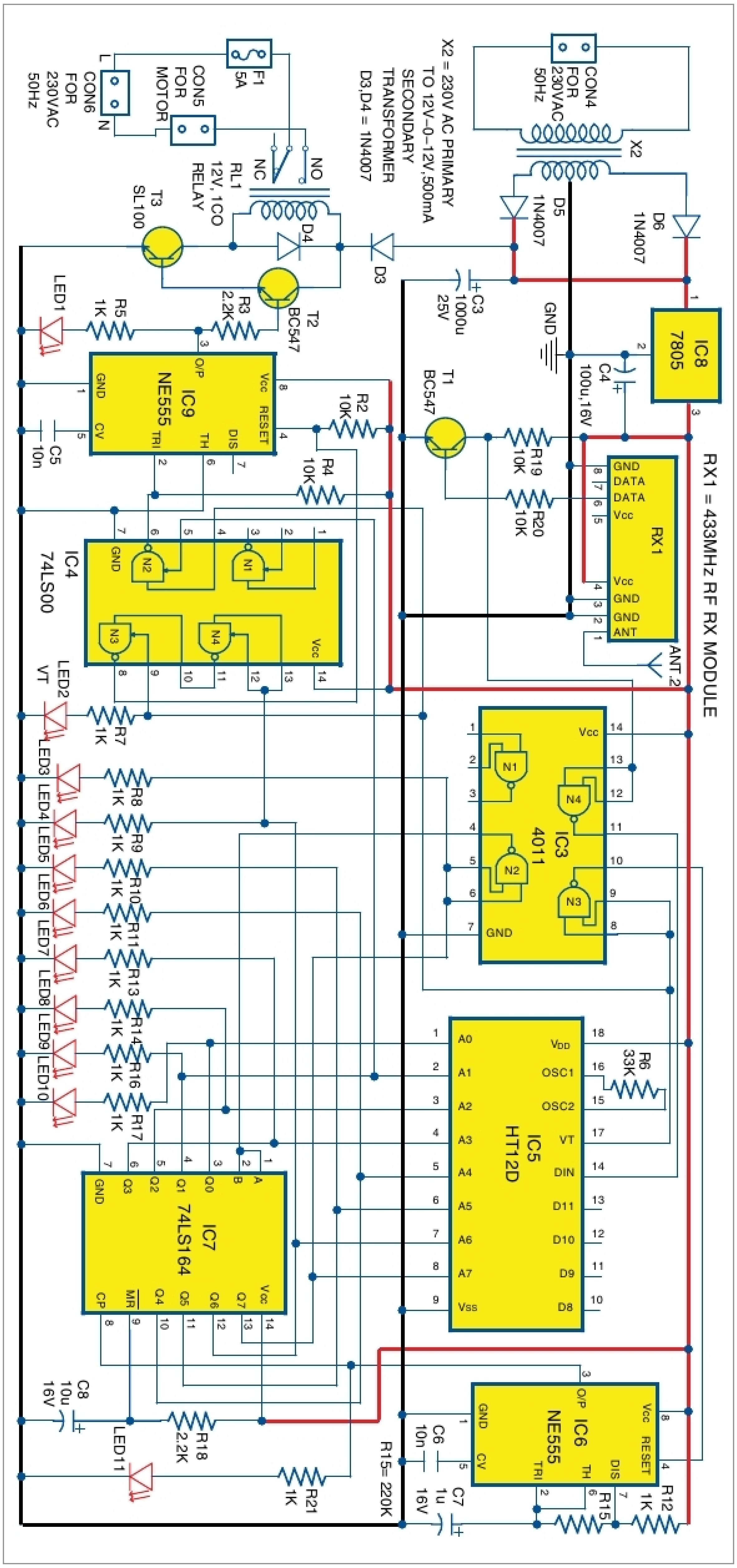

Fig. 5: Circuit diagram of the receiver unit